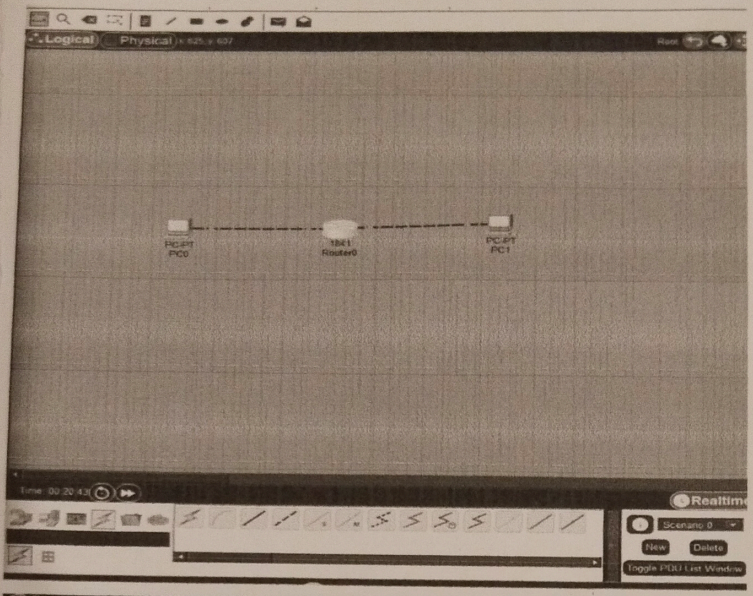


Router0

Physical Config CLI Attributes

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    down
FastEthernet0/1        192.168.20.1    YES manual up                    down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Copy Paste

Exp: 10 Internetworking with Routers
Date: 12/9/24 ~~Cisco Packet Tracer~~

(A) Design and configure a simple Internetwork using a Router.

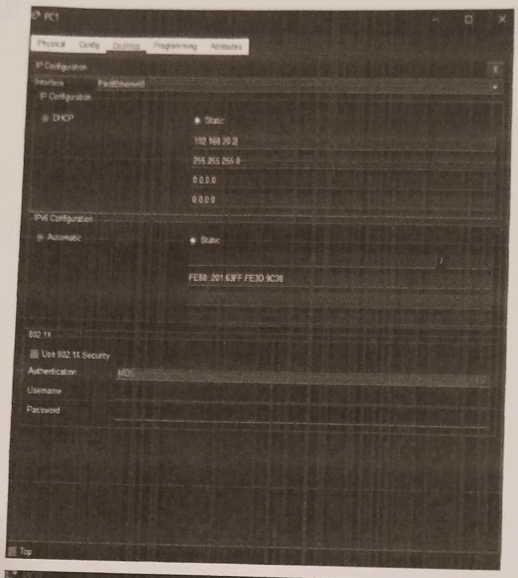
Aim.

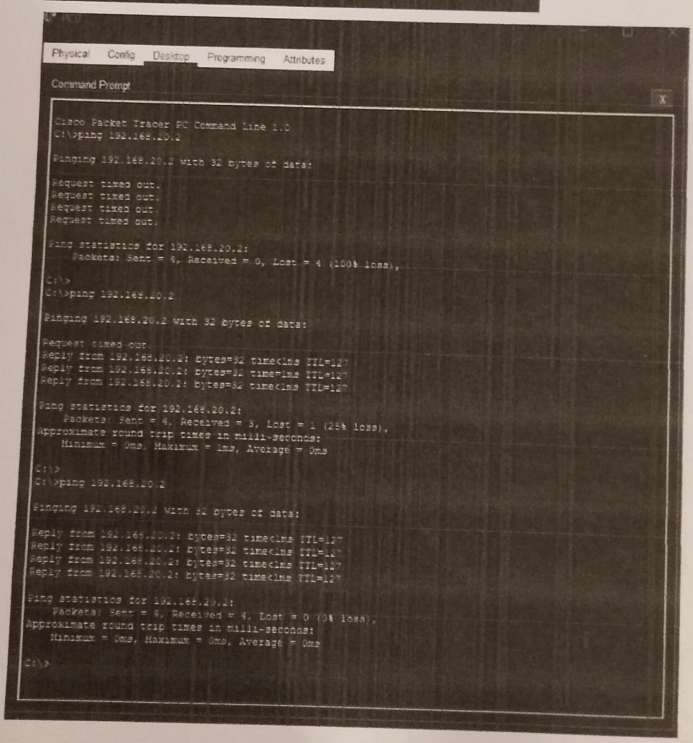
To design and configure an Internetwork consisting of multiple network connected through a router and observe packet transmission b/w different network.

Algorithm :-

1. Open Cisco packet Tracer.
2. Place a router, switches using copper straight through cables.
3. Connect the PCs to switches using copper straight through cables.
4. Connect each switch to a separate router interface.
5. Assign static IP addresses and subnet masking to all PCs
6. Configure the router interfaces with IP addresses belonging to their respective network.
7. Enable the router interfaces using the no shutdown command.
8. Configure the default gateway on each PC using the IP address of the local router interface.

10


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
192.168.20.2
255.255.255.0
0.0.0.0
0.0.0.0
IPv6 Configuration
Automatic
Static
FE80::201:63FF:FE30:9C38
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

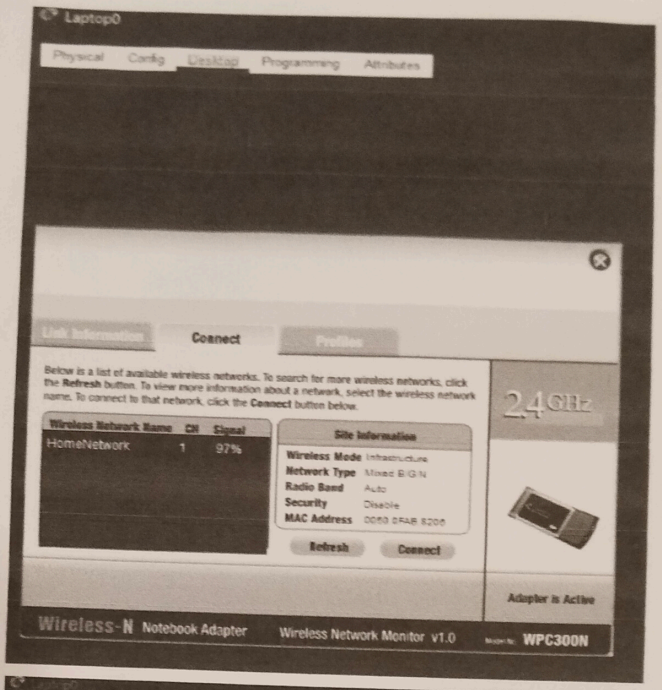

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

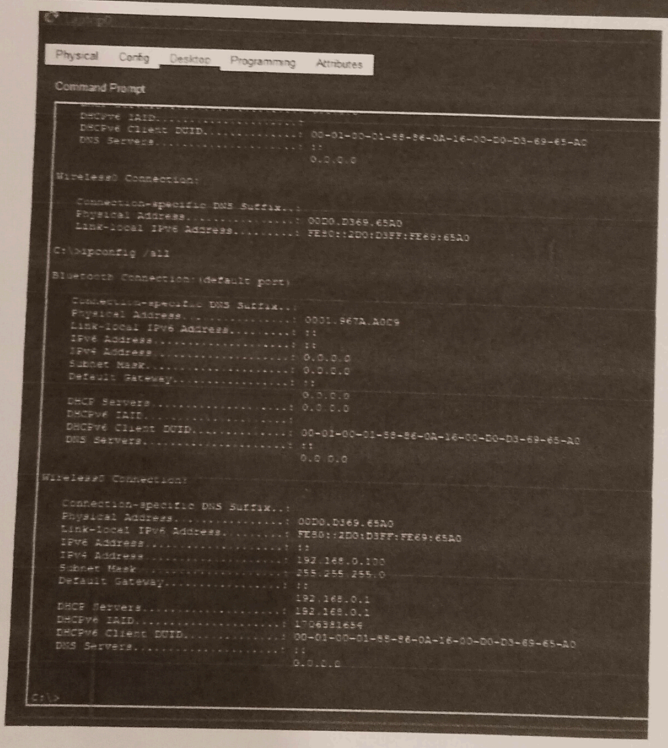
9. Test connectivity b/w PCs in different networks using the ping command.

10. Use simulation mode to observe how packets travel through the router.

Result:

A simple Internetwork was designed and configured successfully. Communication b/w different network was verified and packet forwarding through the router was observed.


Laptop0
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
HomeNetwork
1
97%
Site Information
Wireless Mode
Infrastructure
Network Type
Mixed B/G/N
Radio Band
Auto
Security
Disable
MAC Address
0060 5FAB 8206
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N


Physical
Config
Desktop
Programming
Attributes
Command Prompt
DHCPv6 IAID.....
DHCPv6 Client DUID.....: 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Physical Address.....: 00D0.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
C:\>ipconfig /all
Bluetooth Connection:(default port)
Connection-specific DNS Suffix..:
Physical Address.....: 0031.967A.A0C9
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....
DHCPv6 Client DUID.....: 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Physical Address.....: 00D0.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.0.1
DHCP Servers.....: 192.168.0.1
DHCPv6 IAID.....: 170681684
DHCPv6 Client DUID.....: 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
C:\>

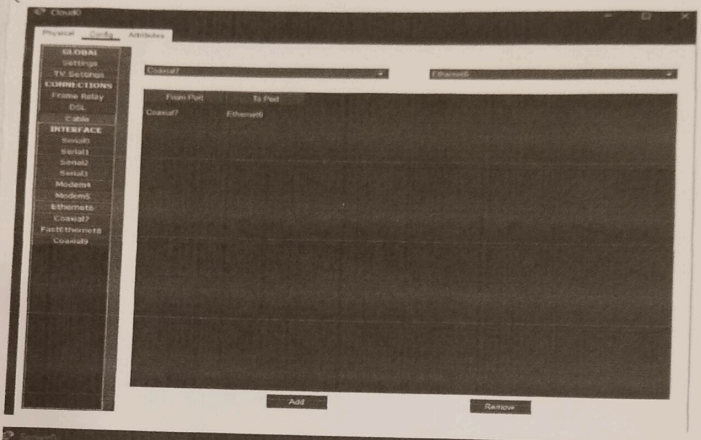
6) Internetwork using wireless Router, DHCP Server and Internet cloud.

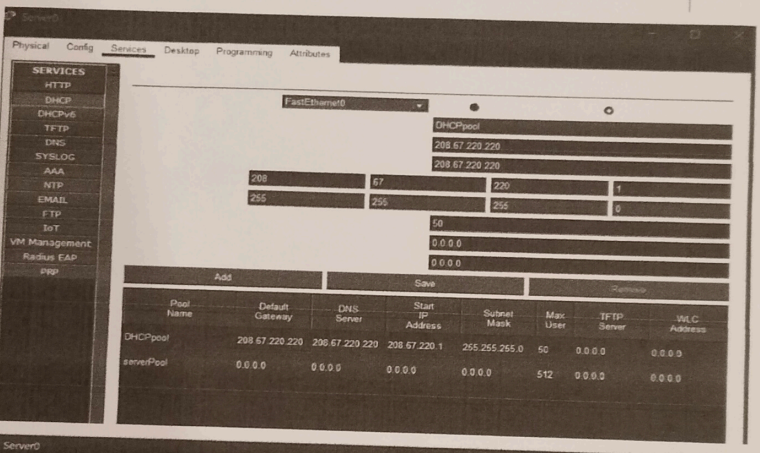
Aim:-

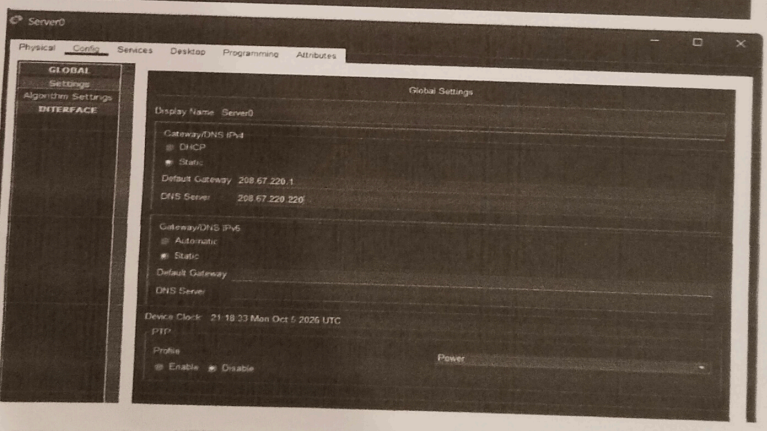
To configure an Internetwork using a wireless router, DHCP server and internet cloud in Cisco packet tracer.

Algorithm:-

1. Open Cisco packet tracer and place a wireless router, wireless and devices a server and an Internet cloud.
2. Configure the wireless router with an SSID and Security settings.
3. Enable DHCP and Configure the IP address pool, Subnet mask, default gateway, and DNS server if required.
4. Configure the server with a suitable static IP address and enable the required services.
5. Connect the router to the Internet cloud using the appropriate interface and connection.
6. Connect wireless devices to the configured SSID.
7. Verify that the devices receive IP addresses automatically through DHCP


Cloud0
Physical
Config
Attributes
GLOBAL
Settings
IPv Settings
CONNECTIONS
Frame Relay
DSL
Cable
INTERFACE
Serial0
Serial1
Serial2
Serial3
Modem4
Modem5
Ethernet6
Coaxial7
FastEthernet8
Coaxial9
Coaxial7
Ethernet6
From Port
To Port
Coaxial7
Ethernet6
Add
Remove

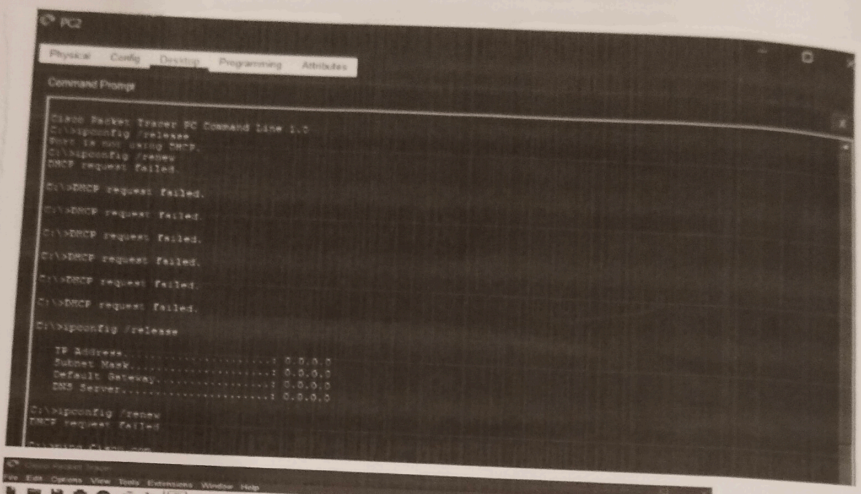

Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PPP
FastEthernet0
DHCPpool
208.67.220.220
208.67.220.220
208
67
220
4
255
255
255
0
50
0.0.0.0
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
DHCPpool
208.67.220.220
208.67.220.220
208.67.220.1
255.255.255.0
50
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
0.0.0.0
0.0.0.0
512
0.0.0.0
0.0.0.0

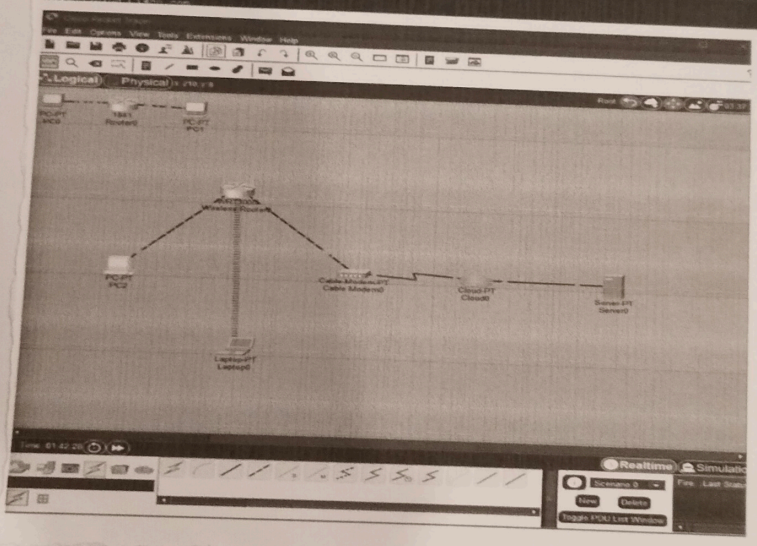

Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Global Settings
Display Name Server0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 208.67.220.1
DNS Server 208.67.220.220
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock 21:18:33 Mon Oct 5 2026 UTC
PTP
Profile
Power
Enable
Disable

8. Test connectivity to the server and other reachable network devices.

Result:-

The wireless network was configured successfully devices obtained IP addresses automatically through DHCP, and connectivity was tested.


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /release
Host is not using DHCP.
C:\>ipconfig /renew
DHCP request failed
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>ipconfig /release
IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0
DNS Server.....: 0.0.0.0
C:\>ipconfig /renew
DHCP request failed


Logical
Physical
Wireless Router0
PC-PT PC2
Cable Modem-PT Cable Modem0
Cloud-PT Cloud0
Server-PT Server0
Laptop-PT Laptop0
Realtime
Simulation
New
Delete
Toggle PDU List Window

(e) Internetwork using switches, a router, and Ethernet cables.

Aim:-

To design and configure an Internetwork using switches, a router and Ethernet cables in the laboratory.

Algorithm:-

1. Place the required PCs, switches and router in Cisco packet tracer or Arrange the devices in the laboratory.
2. Connect PCs to switches using Ethernet cables.
3. Connect the switches to the router using suitable Ethernet cables.
4. Assign static IP addresses and subnet masks to all PCs
5. Configure the router interface with appropriate IP addresses
6. Enable the router interfaces.
7. Configure the default gateway on each PC
8. Test communication between devices in the same and different network using ping.
9. Observe the packet transmissions & verify connectivity.

Result:-

The Internetwork was configured successfully using switches a router and Ethernet cables Connectivity b/w the configured network was verified.